

B30801 Managerial Accounting Autumn 2025

Doug Skinner

RegionAirFly

RegionAir

SAAB 2000



-
- What was RegionFly's original business model? What happened to change this? Is it still viable?
 - What was the focus of the key decision-makers after the acquisition? (What to maximize?)
 - Should we drop Route 7 in 2019?
 - Need to understand cost behavior.
 - How to determine which costs are fixed, variable?
 - What non-financial considerations should we consider, if any?



Baldwin King,
RegionFly
founder

“By investing resources in unnecessary growth rather than focusing on our core values of customer service and flight quality, RegionFly would stray from its original business model and ultimately lose its dominant position in the Southeastern airline niche market”

Cost systems

Cost systems perform two functions:

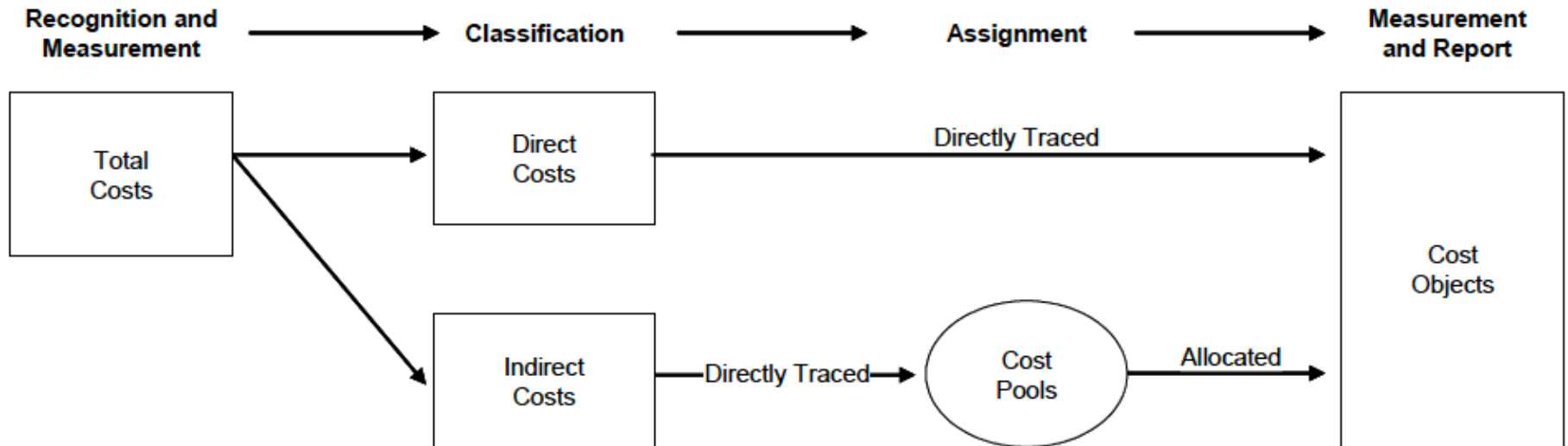
- **Accumulate cost data:** as resources are acquired and paid for, the accounting system captures the associated costs (at some point there is a cash outflow).

Cost systems

Cost systems perform two functions:

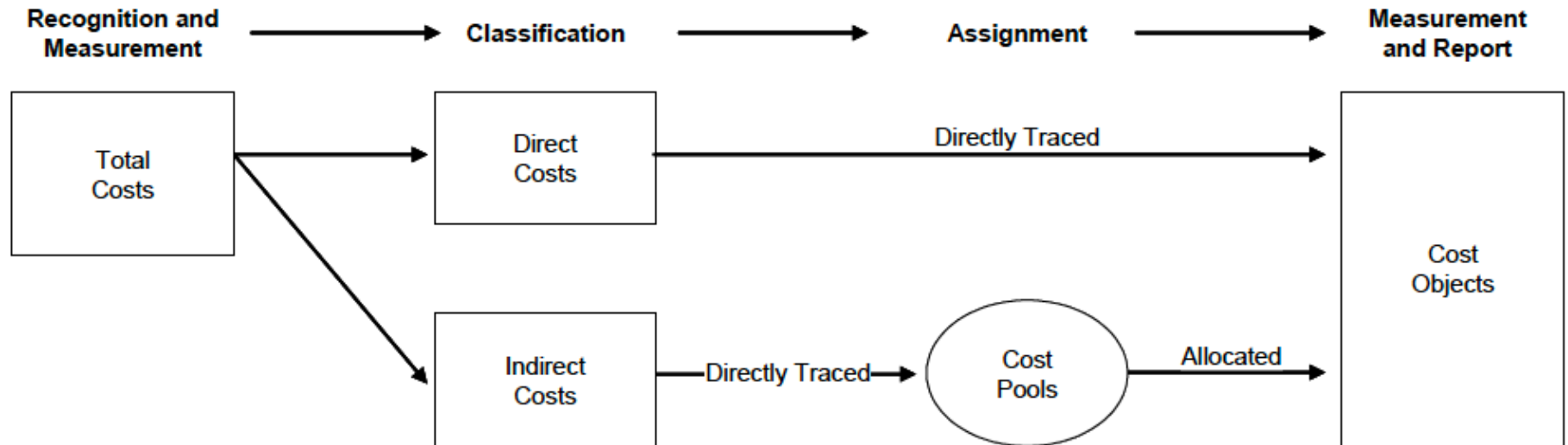
- **Accumulate cost data:** as resources are acquired and paid for, the accounting system captures the associated costs (at some point there is a cash outflow).
- **Assign costs,** to the activities, products, and services that drive value creation (we will refer to these as **cost objects**).
 - For RegionFly, the cost objects are the routes.
 - This is done either **directly** (direct costs) or by way of an **allocation rule** (indirect costs).

Basic structure of Cost Systems.



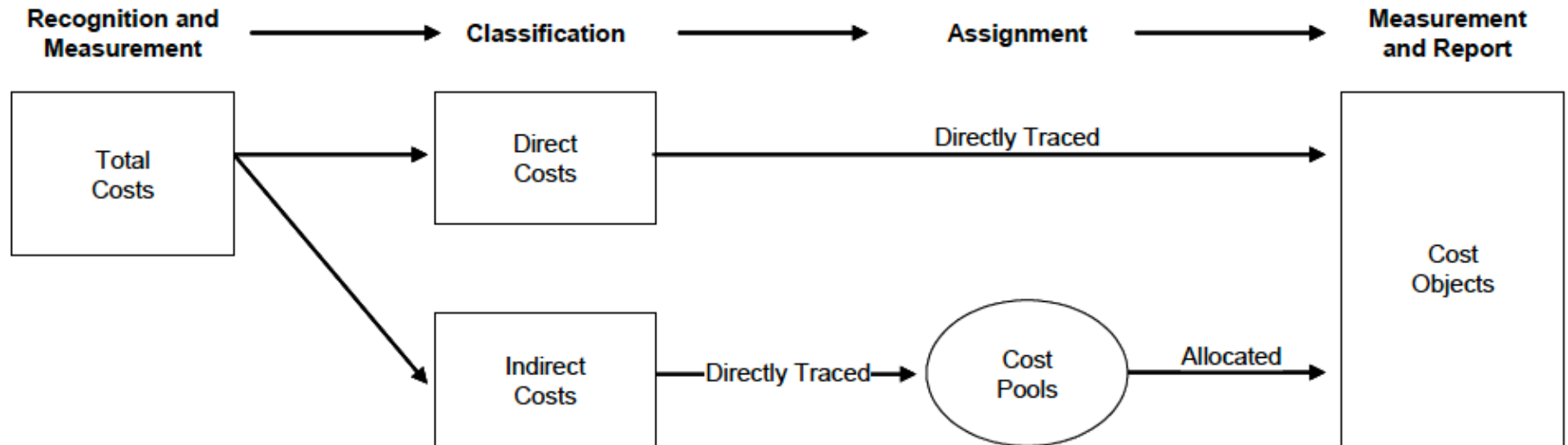
Source: Introduction to Cost-Accounting Systems (HBS).

Basic structure of Cost Systems.



Whether and how to allocate indirect costs to cost objects is always purpose-dependent – we don't always allocate indirect costs and may do so in different ways for different purposes (e.g., performance evaluation versus control versus resource allocation/investment).

Basic structure of Cost Systems.



Full costing – all costs (direct and indirect, fixed and variable) are allocated to products.

Cost Allocation

Cost pools: buckets of accumulated costs we may assign to cost objects:

- For RegionFly, these are **variable direct costs** (directly traced to routes; see Ex. 1) and **overhead costs**, accumulated in the different accounts (bottom of Ex. 1).
- The overhead (indirect) costs must be *allocated* from the cost pools to the routes in some way.

Exhibit 1 RegionFly Revenue, Cost, and Profitability, 2015–2018

	2015	2016	2017	2018
REVENUE				
Route #				
Route #1	\$1,117,285	\$1,048,797	\$1,278,508	\$1,273,941
Route #2	\$833,450	\$830,014	\$0	\$0
Route #3	\$1,223,577	\$1,153,819	\$1,100,879	\$901,845
Route #4	\$556,235	\$523,553	\$0	\$0
Route #5	\$890,605	\$808,126	\$619,590	\$608,818
Route #6	\$491,897	\$473,232	\$548,132	\$539,726
Route #7	\$889,480	\$875,692	\$869,813	\$840,010
Total Revenue	\$6,002,529	\$5,713,233	\$4,416,922	\$4,164,340
VARIABLE DIRECT COSTS				
Route #				
Route #1	\$173,051	\$167,593	\$213,085	\$205,474
Route #2	\$170,589	\$195,632	\$0	\$0
Route #3	\$195,612	\$210,962	\$195,643	\$154,203
Route #4	\$104,711	\$107,596	\$0	\$0
Route #5	\$122,443	\$130,971	\$99,896	\$98,196
Route #6	\$78,872	\$79,338	\$91,589	\$84,332
Route #7	\$168,453	\$161,250	\$171,569	\$180,000
Total Variable Direct Costs	\$1,013,732	\$1,053,342	\$771,782	\$722,206
OVERHEAD				
Account #				
1000	\$755,242	\$787,084	\$576,695	\$538,631
2000	\$134,865	\$138,900	\$117,983	\$111,485
3000	\$485,512	\$510,102	\$373,751	\$349,082
4000	\$755,242	\$783,004	\$761,080	\$717,969
5000	\$269,729	\$281,473	\$264,022	\$242,362
6000	\$53,946	\$56,287	\$53,585	\$47,592
7000	\$242,756	\$252,165	\$184,761	\$172,566
Total Overhead Costs	\$2,697,292	\$2,809,015	\$2,331,876	\$2,179,686
Operating Profit	\$2,291,506	\$1,850,877	\$1,313,264	\$1,262,448

	2015	2016	2017	2018
REVENUE				
Route #				
Route #1	\$1,117,285	\$1,048,797	\$1,278,508	\$1,273,941
Route #2	\$833,450	\$830,014	\$0	\$0
Route #3	\$1,223,577	\$1,153,819	\$1,100,879	\$901,845
Route #4	\$556,235	\$523,553	\$0	\$0
Route #5	\$890,605	\$808,126	\$619,590	\$608,818
Route #6	\$491,897	\$473,232	\$548,132	\$539,726
Route #7	\$889,480	\$875,692	\$869,813	\$840,010
Total Revenue	\$6,002,529	\$5,713,233	\$4,416,922	\$4,164,340

-5% (declines all routes)

VARIABLE DIRECT COSTS				
Route #				
Route #1	\$173,051	\$167,593	\$213,085	\$205,474
Route #2	\$170,589	\$195,632	\$0	\$0
Route #3	\$195,612	\$210,962	\$195,643	\$154,203
Route #4	\$104,711	\$107,596	\$0	\$0
Route #5	\$122,443	\$130,971	\$99,896	\$98,196
Route #6	\$78,872	\$79,338	\$91,589	\$84,332
Route #7	\$168,453	\$161,250	\$171,569	\$180,000
Total Variable Direct Costs	\$1,013,732	\$1,053,342	\$771,782	\$722,206

+4% (increases for 5 routes)

OVERHEAD				
Account #				
1000	\$755,242	\$787,084	\$576,695	\$538,631
2000	\$134,865	\$138,900	\$117,983	\$111,485
3000	\$485,512	\$510,102	\$373,751	\$349,082
4000	\$755,242	\$783,004	\$761,080	\$717,969
5000	\$269,729	\$281,473	\$264,022	\$242,362
6000	\$53,946	\$56,287	\$53,585	\$47,592
7000	\$242,756	\$252,165	\$184,761	\$172,566
Total Overhead Costs	\$2,697,292	\$2,809,015	\$2,331,876	\$2,179,686

Operating Profit	\$2,291,506	\$1,850,877	\$1,313,264	\$1,262,448
------------------	-------------	-------------	-------------	-------------

-19%

What happens in 2016?

+4% overall in 2016
and increase for all
categories

	2015	2016	2017	2018
REVENUE				
Route #				
Route #1	\$1,117,285	\$1,048,797	\$1,278,508	\$1,273,941
Route #2	\$833,450	\$830,014	\$0	\$0
Route #3	\$1,223,577	\$1,153,819	\$1,100,879	\$901,845
Route #4	\$556,235	\$523,553	\$0	\$0
Route #5	\$890,605	\$808,126	\$619,590	\$608,818
Route #6	\$491,897	\$473,232	\$548,132	\$539,726
Route #7	\$889,480	\$875,692	\$869,813	\$840,010
Total Revenue	\$6,002,529	\$5,713,233	\$4,416,922	\$4,164,340

-5%

VARIABLE DIRECT COSTS				
Route #				
Route #1	\$173,051	\$167,593	\$213,085	\$205,474
Route #2	\$170,589	\$195,632	\$0	\$0
Route #3	\$195,612	\$210,962	\$195,643	\$154,203
Route #4	\$104,711	\$107,596	\$0	\$0
Route #5	\$122,443	\$130,971	\$99,896	\$98,196
Route #6	\$78,872	\$79,338	\$91,589	\$84,332
Route #7	\$168,453	\$161,250	\$171,569	\$180,000
Total Variable Direct Costs	\$1,013,732	\$1,053,342	\$771,782	\$722,206

+4%

OVERHEAD				
Account #				
1000	\$755,242	\$787,084	\$576,695	\$538,631
2000	\$134,865	\$138,900	\$117,983	\$111,485
3000	\$485,512	\$510,102	\$373,751	\$349,082
4000	\$755,242	\$783,004	\$761,080	\$717,969
5000	\$269,729	\$281,473	\$264,022	\$242,362
6000	\$53,946	\$56,287	\$53,585	\$47,592
7000	\$242,756	\$252,165	\$184,761	\$172,566
Total Overhead Costs	\$2,697,292	\$2,809,015	\$2,331,876	\$2,179,686

-19%

Operating Profit	\$2,291,506	\$1,850,877	\$1,313,264	\$1,262,448
-------------------------	--------------------	--------------------	--------------------	--------------------

What happens in 2016?

Is this a transitory macro/industry shock (like COVID)? Or a systemic change in the industry?

Q2.

Let's drill down a bit to better understand the overhead costs and how they change...

What's fixed vs. variable?

- If activity increases, will costs increase? Will they do so proportionately?

What's fixed vs. variable?

- If activity increases, will costs increase? Will they do so proportionately?

For RegionFly, we know there's a significant drop in activity in 2017.

	2015	2016	2017	2018
OVERHEAD				
Account #				
1000	\$755,242	\$787,084	\$576,695	\$538,631
2000	\$134,865	\$138,900	\$117,983	\$111,485
3000	\$485,512	\$510,102	\$373,751	\$349,082
4000	\$755,242	\$783,004	\$761,080	\$717,969
5000	\$269,729	\$281,473	\$264,022	\$242,362
6000	\$53,946	\$56,287	\$53,585	\$47,592
7000	\$242,756	\$252,165	\$184,761	\$172,566
Total Overhead Costs	\$2,697,292	\$2,809,015	\$2,331,876	\$2,179,686

Exhibit 2 RegionFly Chart of Overhead Accounts

Account #	Account	Description
1000	Flight Support Expenses	Passenger service and flying operating expenses, personnel expenses and fringe benefits for engineers and flight attendants, pilots, co-pilots that are not assigned to a particular route
2000	Sales Expenses	Wages and fringe benefits of reservations personnel and sales agents. Other selling expenses, including ticketing, sales and promotions, company-wide advertising and publicity
3000	Maintenance Expenses	Wages and fringe benefits of repair and maintenance workers, related equipment, materials, oil, and repair expenses
4000	Fees and Taxes	Aircraft financing expenses (interests on capital leases or loans), insurance, operating lease payments, rental costs, taxes, and Federal Aviation Authority registration and airport fees (e.g., landing fees)
5000	Depreciation	Straight-line depreciation on owned aircrafts, maintenance facilities, buildings, and leased equipment (capital leases)
6000	Other Expenses	Administrative costs, including office supplies (i.e., books, charts) and legal costs
7000	Airport Expenses	Wages and fringe benefits of air traffic and ground service personnel, including gate agents, baggage handlers, baggage claim operators, and cargo operators

[illegible]

				2015		2016		2017			2018		
	Overhead by account												
		1000		755,242		787,084		576,695			538,631		
		2000		134,865		138,900		117,983			111,485		
		3000		485,512		510,102		373,751			349,082		
		4000		755,242		783,004		761,080			717,969		
		5000		269,729		281,473		264,022			242,362		
		6000		53,946		56,287		53,585			47,592		
		7000		242,756		252,165		184,761			172,566		
				2,697,292		2,809,015		2,331,877			2,179,687		

Is there a measure of activity (a cost driver) that explains how the different categories of cost vary?

Let's first consider revenues as a measure of activity:

[illegible]

RegionFly uses VDCs are a measure of activity (allocation base):

		2015		2016		2017		2018	
Total Variable Direct Costs		1,013,732		1,053,342		771,782		722,206	
Overhead by account (with % of total variable direct costs in each year).									
1000		755,242	0.745	787,084	0.747	576,695	0.747	538,631	0.746
2000		134,865	0.133	138,900	0.132	117,983	0.153	111,485	0.154
3000		485,512	0.479	510,102	0.484	373,751	0.484	349,082	0.483
4000		755,242	0.745	783,004	0.743	761,080	0.986	717,969	0.994
5000		269,729	0.266	281,473	0.267	264,022	0.342	242,362	0.336
6000		53,946	0.053	56,287	0.053	53,585	0.069	47,592	0.066
7000		242,756	0.239	252,165	0.239	184,761	0.239	172,566	0.239
		2,697,292	2.661	2,809,015	2.667	2,331,877	3.021	2,179,687	3.018

Shows costs in each category as a fraction of total VDCs

[illegible]

[illegible]

Cost Allocation

Cost allocation rule: the mechanism we use to assign costs in a given cost pool to the cost object. Also known as the **burden rate** or **allocation rate**.

Cost Allocation

Cost allocation rule: the mechanism we use to assign costs in a given cost pool to the cost object. Also known as the **burden rate** or **allocation rate**.

→ For RegionFly, the company has chosen VDCs as the activity measure used to compute the allocation rate. For 2016:

$$\begin{aligned}\text{Rate} &= \text{Total Costs in Pool/activity measure} \\ &= \$2,809,015/\$1,053,342 \\ &= \$2.667\end{aligned}$$

Cost Allocation

Cost allocation rule: the mechanism we use to assign costs in a given cost pool to the cost object. Also known as the **burden rate** or **allocation rate**.

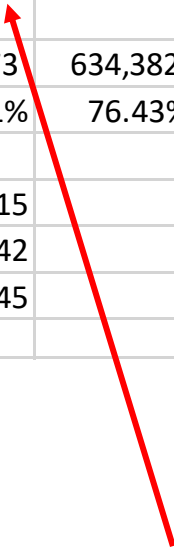
→ For RegionFly, the company has chosen VDCs as the activity measure used to compute the allocation rate. For 2016:

$$\begin{aligned}\text{Rate} &= \text{Total Costs in Pool/activity measure} \\ &= \$2,809,015/\$1,053,342 \\ &= \$2.667\end{aligned}$$

For each route, we allocate \$2.667 of overhead for every dollar of VDC.

			Route 1	Route 2	Route 3	Route 4	Route 5	Route 6	Route 7	Total
	Revenues		1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233
	Variable Direct Costs		(167,593)	(195,632)	(210,962)	(107,596)	(130,971)	(79,338)	(161,250)	(1,053,342)
	Allocated Overhead Costs									
	Profit Margin		881,204	634,382	942,857	415,957	677,155	393,894	714,442	4,659,891
	% Margin		84.02%	76.43%	81.72%	79.45%	83.79%	83.23%	81.59%	81.56%
	Total Overhead		2,809,015							
	Total VDC		1,053,342							
	Allocation rate =		2.6667645							

		2016 Overhead Cost Allocation/Gross Margins										
			Route 1	Route 2	Route 3	Route 4	Route 5	Route 6	Route 7	Total		
		Revenues	1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233		
		Variable Direct Costs	(167,593)	(195,632)	(210,962)	(107,596)	(130,971)	(79,338)	(161,250)	(1,053,342)		
		Allocated Overhead Costs	(446,931)									
		Profit Margin	434,273	634,382	942,857	415,957	677,155	393,894	714,442	4,659,891		
		% Margin	41.41%	76.43%	81.72%	79.45%	83.79%	83.23%	81.59%	81.56%		
		Total Overhead	2,809,015									
		Total VDC	1,053,342									
		Allocation rate =	2.6667645									



$$= 2.6667645 * \$167,593$$

		2016 Overhead Cost Allocation/Gross Margins										
			Route 1	Route 2	Route 3	Route 4	Route 5	Route 6	Route 7	Total		
		Revenues	1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233		
		Variable Direct Costs	(167,593)	(195,632)	(210,962)	(107,596)	(130,971)	(79,338)	(161,250)	(1,053,342)		
		Allocated Overhead Costs	(446,931)									
		Profit Margin	434,273	634,382	942,857	415,957	677,155	393,894	714,442	4,659,891		
		% Margin	41.41%	76.43%	81.72%	79.45%	83.79%	83.23%	81.59%	81.56%		
		Total Overhead	2,809,015									
		Total VDC	1,053,342									
		Allocation rate =	2.6667645									

This is equivalent to allocating overhead costs based on each route's share of total variable direct costs (i.e., 15.9% for Route 1). This is how Ex. 3 shows the calculation.

[illegible]

			2015			2016	
Route 1							
	Revenue		1,117,285			1,048,797	
	Variable Direct Costs		173,051	0.155		167,593	0.160
Route 2							
	Revenue		833,450			830,014	
	Variable Direct Costs		170,589	0.205		195,632	0.236
Route 3							
	Revenue		1,223,577			1,153,819	
	Variable Direct Costs		195,612	0.160		210,962	0.183
Route 4							
	Revenue		556,235			523,553	
	Variable Direct Costs		104,711	0.188		107,596	0.206
Route 5							
	Revenue		890,605			808,126	
	Variable Direct Costs		122,443	0.137		130,971	0.162
Route 6							
	Revenue		491,897			473,232	
	Variable Direct Costs		78,872	0.160		79,338	0.168
Route 7							
	Revenue		889,480			875,692	
	Variable Direct Costs		168,453	0.189		161,250	0.184

Routes 2 and 4 show the lowest margins because they have the highest % VDCs in 2016 (they don't look that much different in 2015).

Why? Company cut prices on these routes to stay competitive.

Because of this, these routes are assigned relatively more overhead costs.

		2016 Overhead Cost Allocation/Gross Margins										
			Route 1	Route 2	Route 3	Route 4	Route 5	Route 6	Route 7	Total		
		Revenues	1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233		
		Variable Direct Costs	(167,593)	(195,632)	(210,962)	(107,596)	(130,971)	(79,338)	(161,250)	(1,053,342)		
		Allocated Overhead Costs	(446,931)	(521,704)	(562,586)	(286,933)	(349,269)	(211,576)	(430,016)	(2,809,015)		
		Profit Margin	434,273	112,678	380,271	129,024	327,886	182,318	284,426	1,850,876		
		% Margin	41.41%	13.58%	32.96%	24.64%	40.57%	38.53%	32.48%	32.40%		
		Total Overhead	2,809,015									
		Total VDC	1,053,342									
		Allocation rate =	2.6667645									

What is consultants' logic for dropping Routes 2 and 4?

2016 Overhead Cost Allocation/Gross Margins										
			Route 1	Route 2	Route 3	Route 4	Route 5	Route 6	Route 7	Total
	Revenues	1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233	
	Variable Direct Costs	(167,593)	(195,632)	(210,962)	(107,596)	(130,971)	(79,338)	(161,250)	(1,053,342)	
	Allocated Overhead Costs	(446,931)	(521,704)	(562,586)	(286,933)	(349,269)	(211,576)	(430,016)	(2,809,015)	
	Profit Margin	434,273	112,678	380,271	129,024	327,886	182,318	284,426	1,850,876	
	% Margin	41.41%	13.58%	32.96%	24.64%	40.57%	38.53%	32.48%	32.40%	
	Total Overhead	2,809,015								
	Total VDC	1,053,342								
	Allocation rate =	2.6667645								

The differences in % margins across routes are more pronounced than before allocating overhead costs.

The overall margin is 32.4% (a weighted average). The consultants' logic is that by dropping routes 2 and 4, this will increase. Is this correct?

What happens in 2017 after we cut capacity?
Let's look at how the overhead and overhead allocation changes.

Q3. Overhead Allocation Rates						
		2015	2016	2017	2018	
	Variable Direct Costs	1,013,732	1,053,342	771,782	722,206	
	Total Overhead Costs	2,697,292	2,809,015	2,331,876	2,179,686	
	Allocation Rate	2.661	2.667	3.021	3.018	

Why is the rate higher in 2017 and 2018?

		Q3. Overhead Allocation Rates					
			2015	2016	2017	2018	
		Variable Direct Costs	1,013,732	1,053,342	771,782	722,206	
		Total Overhead Costs	2,697,292	2,809,015	2,331,876	2,179,686	
		Allocation Rate	2.661	2.667	3.021	3.018	
		The rate increases in 2017 and 2018 relative to prior years because overhead costs do not decline as much as the variable direct costs used as the base (because many of the overhead costs are fixed or partly fixed) so the numerator declines more slowly than the denominator.					
		Decline in VDC			-26.7%		
		Decline in overhead costs			-17.0%		
		Decline in revenues			-22.7%		

Are we better off in 2017?

		2017 Overhead Cost Allocation/Gross Margins						
			Route 1	Route 3	Route 5	Route 6	Route 7	Total
		Revenues	1,278,508	1,100,879	619,590	548,132	869,813	4,416,922
		Variable Direct Costs	(213,085)	(195,643)	(99,896)	(91,589)	(171,569)	(771,782)
		Allocated Overhead Costs	(643,819)	(591,119)	(301,828)	(276,729)	(518,382)	(2,331,876)
		Profit Margin	421,604	314,117	217,866	179,814	179,862	1,313,264
		%	32.98%	28.53%	35.16%	32.80%	20.68%	29.73%
		<i>2016 Comparables</i>	<i>41.41%</i>	<i>32.96%</i>	<i>40.57%</i>	<i>38.53%</i>	<i>32.48%</i>	<i>1,850,876</i>
								<i>32.40%</i>
		Total Overhead	2,331,876					
		Total VDC	771,782					
		Allocation rate =	3.021418					

		2017 Overhead Cost Allocation/Gross Margins						
			Route 1	Route 3	Route 5	Route 6	Route 7	Total
		Revenues	1,278,508	1,100,879	619,590	548,132	869,813	4,416,922
		Variable Direct Costs	(213,085)	(195,643)	(99,896)	(91,589)	(171,569)	(771,782)
		Allocated Overhead Costs	(643,819)	(591,119)	(301,828)	(276,729)	(518,382)	(2,331,876)
		Profit Margin	421,604	314,117	217,866	179,814	179,862	1,313,264
		%	32.98%	28.53%	35.16%	32.80%	20.68%	29.73%
		<i>2016 Comparables</i>	<i>41.41%</i>	<i>32.96%</i>	<i>40.57%</i>	<i>38.53%</i>	<i>32.48%</i>	<i>1,850,876</i>
								<i>32.40%</i>
		Total Overhead	2,331,876					
		Total VDC	771,782					
		Allocation rate =	3.021418					

When we dropped routes, revenues fell more than costs (some costs are fixed or semivariable). Moreover, because allocation rate increased, total \$ of overhead allocated to fewer routes, making them *look* less profitable.

What was the contribution margin of the two routes (ignoring overhead costs)?

			2015				2016	
		Route 2	Route 4	Total		Route 2	Route 4	Total
Revenues		833,450	556,235			830,014	523,553	
VDCs		(170,589)	(104,711)			(195,632)	(107,596)	
CM		662,861	451,524	1,114,385		634,382	415,957	1,050,339

Thinking about 2019.

The first (benchmark) scenario is do nothing – keep route 7 – in which case 2019 is assumed to look the same as 2018.

If we drop route 7, we know for sure route 7's revenues and VDCs go away; the question is what happens to overhead costs.

How do we model changes in overhead costs?

I explored three scenarios based on different assumptions:

- Remaining overhead costs are fully fixed and don't decline further.
- Remaining overhead costs drop by 17%, as they did in 2017 (there is a similar % drop in VDCs).
- Treat 1000, 3000, and 7000 as variable, so that they decline proportionately to VDCs while the rest are fixed.

2015			2016			2017			2018			2019F (b3)	
1,013,731			1,053,342			771,782			722,205			542,206	
755,242	0.745		787,084	0.747		576,695	0.747		538,631	0.746		406,655	0.75
134,865	0.133		138,900	0.132		117,983	0.153		111,485	0.154		111,485	
485,512	0.479		510,102	0.484		373,751	0.484		349,082	0.483		260,259	0.48
755,242	0.745		783,004	0.743		761,080	0.986		717,969	0.994		717,969	
269,729	0.266		281,473	0.267		264,022	0.342		242,362	0.336		242,362	
53,946	0.053		56,287	0.053		53,585	0.069		47,592	0.066		47,592	
242,756	0.239		252,165	0.239		184,761	0.239		172,566	0.239		130,129	0.24
2,697,292	2.661		2,809,015	2.667		2,331,877	3.021		2,179,687	3.018		1,738,729	

Q4.									
		2019F (a)		2019F (b1)		2019F (b2)		2019F (b3)	
	Revenue	4,164,340		3,324,330		3,324,330		3,324,330	
	Variable direct costs	722,206	0.173	542,205	0.163	542,205	0.163	542,205	0.163
	Overhead costs	2,179,686	0.523	2,179,686	0.656	1,809,139	0.544	1,738,729	0.523
	Total costs	2,901,892	0.697	2,721,891	0.819	2,351,344	0.707	2,280,934	0.686
	Pre-tax profit	1,262,448	0.303	602,439	0.181	972,986	0.293	1,043,396	0.314
	Overhead rate	3.02		4.02		3.34		3.21	

- a. Keep Route 7 (nothing changes from 2018). Maintain status quo.
- b1. Drop Route 7 (assume overhead costs all fixed).
- b2. Drop Route 7 (assume overhead costs decline by 17%, as they did in 2017)
- b3. Drop Route 7 (treat 1000, 3000, 7000 as variable; rest as fixed)

Q4.									
		2019F (a)		2019F (b1)		2019F (b2)		2019F (b3)	
	Revenue	4,164,340		3,324,330		3,324,330		3,324,330	
	Variable direct costs	722,206	0.173	542,205	0.163	542,205	0.163	542,205	0.163
	Overhead costs	2,179,686	0.523	2,179,686	0.656	1,809,139	0.544	1,738,729	0.523
	Total costs	2,901,892	0.697	2,721,891	0.819	2,351,344	0.707	2,280,934	0.686
	Pre-tax profit	1,262,448	0.303	602,439	0.181	972,986	0.293	1,043,396	0.314
	Overhead rate	3.02		4.02		3.34		3.21	

- a. Keep Route 7 (nothing changes from 2018). Maintain status quo.
- b1. Drop Route 7 (assume overhead costs all fixed).
- b2. Drop Route 7 (assume overhead costs decline by 17%, as they did in 2017)
- b3. Drop Route 7 (treat 1000, 3000, 7000 as variable; rest as fixed)

Q4.									
		2019F (a)		2019F (b1)		2019F (b2)		2019F (b3)	
	Revenue	4,164,340		3,324,330		3,324,330		3,324,330	
	Variable direct costs	722,206	0.173	542,205	0.163	542,205	0.163	542,205	0.163
	Overhead costs	2,179,686	0.523	2,179,686	0.656	1,809,139	0.544	1,738,729	0.523
	Total costs	2,901,892	0.697	2,721,891	0.819	2,351,344	0.707	2,280,934	0.686
	Pre-tax profit	1,262,448	0.303	602,439	0.181	972,986	0.293	1,043,396	0.314
	Overhead rate	3.02		4.02		3.34		3.21	

Should keep Route 7 if we want to maximize \$ of profit. Overhead rates all go up if we drop Route 7 for the same reasons as previously.

Q4.									
		2019F (a)		2019F (b1)		2019F (b2)		2019F (b3)	
	Revenue	4,164,340		3,324,330		3,324,330		3,324,330	
	Variable direct costs	722,206	0.173	542,205	0.163	542,205	0.163	542,205	0.163
	Overhead costs	2,179,686	0.523	2,179,686	0.656	1,809,139	0.544	1,738,729	0.523
	Total costs	2,901,892	0.697	2,721,891	0.819	2,351,344	0.707	2,280,934	0.686
	Pre-tax profit	1,262,448	0.303	602,439	0.181	972,986	0.293	1,043,396	0.314
	Overhead rate	3.02		4.02		3.34		3.21	

Should keep Route 7 if we want to maximize \$ of profit. Overhead rates all go up if we drop Route 7 for the same reasons as previously.

Route 7 generated around \$700K of CM in 2015, 2016, and 2017, and around \$660K of CM in 2018 – why would we want to lose this?

What would you do?

What about non-financial factors?

- Customer satisfaction.
- Employee morale/cooperation with cost-cutting initiative.

Hong Kong / Transport

Hong Kong's Cathay Pacific Airways warns pilots against slow taxiing after some accused of congesting airport

- Internal airline memo seen by Post says some pilots going at 'considerably slower speed' while taxiing
- Union leader says go-slow issues reflect poor morale among pilots as salaries stay at 60 per cent of pre-pandemic levels despite record profits

 Ezra Cheung

+ FOLLOW

Published: 7:55pm, 14 May, 2023 ▾

 [Why you can trust SCMP](#)

Source: South China Morning Post, May 14, 2023.

Contribution Margin Approach												
			2015		2016		2017			2018		
Revenue			6,002,529		5,713,233		4,416,922			4,164,340		
Variable Costs:												
	Variable Direct Costs		1,013,732	0.169	1,053,342	0.184	771,782	0.175		722,206	0.173	
	Flight Support (1000)		755,242	0.126	787,084	0.138	576,695	0.131		538,631	0.129	
	Maintenance (3000)		485,512	0.081	510,102	0.089	373,751	0.085		349,082	0.084	
	Airport expenses (7000)		242,756	0.040	252,165	0.044	184,761	0.042		172,566	0.041	
			2,497,242		2,602,693		1,906,989			1,782,485		
Contribution Margin			3,505,287	0.584	3,110,540	0.544	2,509,933	0.568		2,381,855	0.572	
Non-variable Costs:												
	Sales expenses (2000)		134,865	0.022	138,900	0.024	117,983	0.027		111,485	0.027	
	Financing, fees, taxes (4000)		755,242	0.126	783,004	0.137	761,080	0.172		717,969	0.172	
	Depreciation (5000)		269,729	0.045	281,473	0.049	264,022	0.060		242,362	0.058	
	Admin costs (6000)		53,946	0.009	56,287	0.010	53,585	0.012		47,592	0.011	
			1,213,782		1,259,664		1,196,670			1,119,408		
Pre-tax profit			2,291,505	0.382	1,850,876	0.324	1,313,263	0.297		1,262,447	0.303	

Contribution Margins by Route for 2016											
				1	2	3	4	5	6	7	Total
	Revenues			1,048,797	830,014	1,153,819	523,553	808,126	473,232	875,692	5,713,233
	Variable costs:										
	VDCs			167,593	195,632	210,962	107,596	130,971	79,338	161,250	1,053,342
	1000			125,230	146,181	157,636	80,398	97,865	59,283	120,490	787,084
	3000			81,160	94,739	102,163	52,106	63,425	38,421	78,089	510,102
	7000			40,121	46,833	50,503	25,758	31,354	18,993	38,602	252,165
				414,104	483,385	521,264	265,858	323,615	196,036	398,431	2,602,693
	CM %			634,693	346,629	632,555	257,695	484,511	277,196	477,261	3,110,540
				0.61	0.42	0.55	0.49	0.60	0.59	0.55	0.54
	Indirect non-variable overhead costs										
	2000										138,900
	4000										783,004
	5000										281,473
	6000										56,287
											1,259,664
	Pre-tax profit										1,850,876
	Notes: Variable overhead costs are allocated to each route using VDCs.										

RegionFly: Supply-driven Death Spiral.

- Profitability *perceived* as being too low (falling margins).
- Drop products based on % profit margins calculated after allocating indirect/overhead costs.
- Volumes (revenues) decrease as we drop products.
- Overhead burden rate increases, because overhead costs decline more slowly than allocation base (partially fixed).
- Margins decline further, causing more products to be dropped – revenues decline more than costs.
- Cycle continues until business can no longer cover fixed costs.

RegionFly: Key takeaways

- Conclusion may seem obvious here, but harder to detect in real-world situations with many products, services, and product lines.
- What metric do we ultimately care about?
 - % operating margins or aggregate \$ of contribution to bottom-line? (free cash flow or ROIC).
 - For any entity to survive in the long-run it needs to generate free cash flow.
 - *CFOs focus on EBITDA \$ and cash flow conversion.*